

Pairs — a memory match game

Design report: personas, storyboards, and the choices behind the final prototype.

1. Designer

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- **Course:** SEG3125, Summer 2026

2. Game

Name and type

The game is called **Pairs**. It is a card-matching memory game. The board holds cards in matching pairs, and the player flips two at a time to find every pair while making as few moves as possible.

Pairs has one twist that makes it a real memory test rather than a guessing game: at the start of every round the whole board is revealed for a three-second countdown. The player has to memorize the layout before the cards turn face down and the timer starts.

The player chooses two things before each round:

- **Difficulty:** Easy (6 pairs), Medium (8 pairs), Hard (12 pairs).
- **Card set:** Animals, Fruit, Sports, or Travel.

It also keeps a personal best (fewest moves, then fastest time) for each difficulty and card set.

Inspiration

Source	What I took from it / what I changed
helpfulgame.com — Memory	Confirmed the core mechanic (find matching cards in a grid). Its interface is cramped and plain, with flat blue tiles and no sense of progress. I kept the grid idea but added the memorize peek, a move counter, a timer, clear matched-state feedback, selectable card sets, and a calmer layout.
mathsisfun.com — Simon Says	Showed how a memory game uses difficulty and options to keep one user practising. I borrowed the idea of levels and configuration but chose pair-matching instead of sequence repetition, and gave the setup a cleaner screen.

The game logic (board building, the peek, flipping, matching, timing, scoring) is written from scratch and is not copied from these sites or from tutorial examples.

3. Personas and storyboards

Persona A — Emma Carter

21, undergraduate student. Plays casual games in short gaps between classes.

Intrinsic traits	Competitive, impatient, quick to learn new interfaces.
Relation to technology	High. Lives on her phone, installs and drops apps fast, expects things to be obvious.
Relation to the domain	Plays puzzle and arcade games for fun; treats a memory game as a quick challenge to beat.
Goal	Clear the Hard board and beat her own best time and move count.

Persona B — Robert Clarke

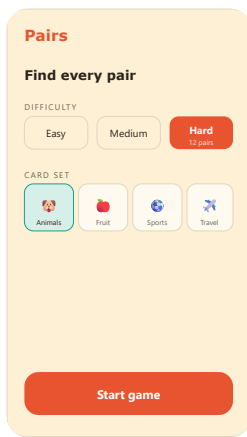
64, retired teacher. Was told that memory games help keep the mind active.

Intrinsic traits	Patient, methodical, careful with unfamiliar apps.
Relation to technology	Moderate. Uses a tablet and a laptop; prefers large text, clear buttons, and no clutter.
Relation to the domain	New to digital games; wants a calm, low-pressure way to practise memory.
Goal	Play a relaxed Easy round in a readable, comfortable interface.

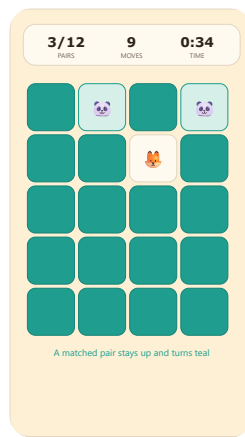
Storyboard A — Emma's run

Warm theme · stats across the top

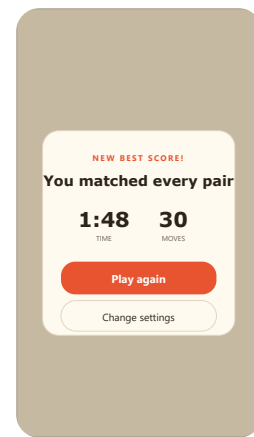
Fast and competitive. Stats sit in one bar across the top so the timer and move count stay in view. She picks Hard, plays at speed, and checks whether she set a new best.



1 · Choose Hard and a card set



2 · Match pairs, stats up top

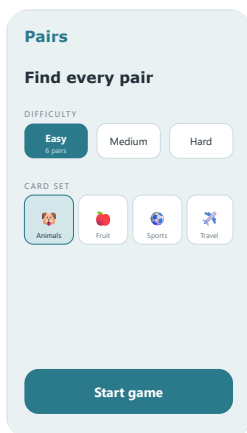


3 · See the result and best score

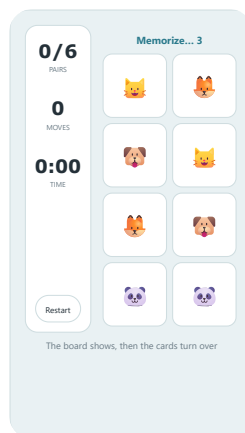
Storyboard B — Robert's run

Cool theme · stats in a side rail

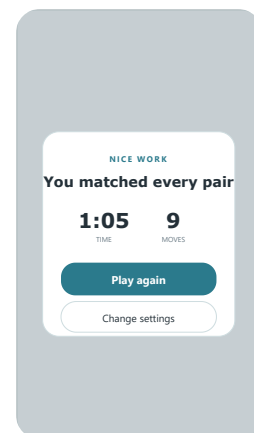
Calm and readable. Larger cards, an Easy board, and the stats moved into a side rail next to the grid. This frame also shows the memorize peek, where the whole board is revealed and counted down before play.



1 · Choose Easy and a card set



2 · Memorize the board, stats at the side



3 · A quiet end screen

How the two storyboards differ: Storyboard A is warm (cream and coral) with the stats in a bar across the top, aimed at a fast, score-driven player. Storyboard B is cool (light blue) with the stats in a side rail and larger cards, aimed at a calm, readable session. They use different colour themes, different layouts for the same information, and show different moments of play (active matching vs. the memorize peek).

4. High-fidelity prototype

Portfolio (public, no login): <https://hadibeiram.github.io/memory-game/> — the game is linked from its Case studies section.

Play the game directly: <https://hadibeiram.github.io/memory-game/#game>

Code: github.com/hadibeiram/memory-game

Relationship to the storyboards

The final prototype takes the warm colour theme and the top stats bar from Storyboard A, because the bar reads cleanly on a phone and keeps the score in view. From Storyboard B it keeps the large, high-contrast cards and the calm, uncluttered feel that Robert needs, and it makes the memorize peek part of every round. The side rail from B was dropped because stacking the stats in one bar works on both a phone and a tablet without a second layout. So the final design is mostly Storyboard A, with the readability and the peek that came out of exploring Storyboard B.

Visual design choices

- **Colour theme.** A warm cream background with one strong coral for the primary action and the selected difficulty, and a teal used for the card backs and for anything "matched". The coral and teal are the only two strong colours, so they carry meaning instead of decorating. A small amount of mustard is used for the focus outline.
- **Typography.** One rounded typeface (Fredoka) is used throughout. Its soft shapes fit a light, playful game, and weight alone (regular for text, bold for the logo, headings, and the large stat numbers) is enough to set the hierarchy.
- **Layout and negative space.** Everything sits in a centred column with room around it, so the board is the obvious focal point. The setup controls live inside one rounded panel, which keeps the start screen calm.
- **Contrast, scale, balance, hierarchy.** Stat numbers are large and bold while their labels are small uppercase text. The single coral button is the clear next step on each screen. Card size and spacing scale together, so the grid stays balanced from phone to desktop.
- **Gestalt principles.**
 - *Similarity:* every face-down card shares the same teal patterned back, so they read as one group; matched cards share a teal "done" state.
 - *Proximity / common region:* the setup controls are grouped inside one bordered panel; the three stats sit together in the bar.
 - *Figure and ground:* cards and panels sit on a plain cream background, and the result dialog dims the board behind it.

- *Continuity and closure*: the aligned grid and the rounded, closed card and panel shapes make the screen easy to scan.

5. Code

GitHub repository: <https://github.com/hadibeiram/memory-game>

Built with React and Vite, deployed to GitHub Pages.

6. Generative AI acknowledgement

I used an AI assistant (Claude) in a limited way: to help draft some of the React and CSS code, sketch the storyboard frames, and produce a first draft of this report. I reviewed and tested everything myself, and the design decisions are my own.